

NOISE AND VIBRATION CONTROL SPECIFICATION FOR MVAC INSTALLATION

NOISE CONTROL SPECIFICATION

PART 1 GENERAL

1.1 Description of Work

1. This section specifies noise control for heating, ventilation & air conditioning installation systems.
2. Provide acoustical performance in accordance with the Contract Documents.
3. Comply with the requirements of Noise Control.
4. In addition to the work covered under this Section, coordinate the equipment supplied under other Sections of the Specifications to comply with the requirements contained herein.
5. The Contractor shall submit to the Project Manager a complete list of all products for use and all related data as indicated in the following section before ordering any materials.
6. The Contractor shall be responsible for the corrective work should the noise levels exceed the requirement set in this specification. The Contractor shall also be responsible for the cost of all tests after such remedial work.

1.2 Work Included

1. Provide complete noise control system as shown or specified, and in accordance with the requirements of the Contract Drawings and Documents.
2. Systems shall be complete with:
 - a) Acoustic linings;
 - b) Sound attenuating units;
 - c) Drywall enclosure (if applicable);
 - d) Acoustic enclosure in terms of metal sheets (if applicable);
 - e) Duct/Pipe lagging (if applicable).

1.3 Quality Assurance

1. Acoustical Criteria:

Noise levels due to equipment, ductwork and pipework shall permit attaining sound pressure levels in all 8 octave bands in occupied spaces conforming to NC curves as per stipulated in the following table. For spaces where NC level requirement is uncertain, it will need to be clarified with Project Manager or project acoustic consultant.

Spaces	NC Target
Conference Rooms	NC 30*
Office type premises: Security Control Room & Management Offices / Network Operation Centre / Office	NC 40*
Office Services Corridor	NC 40*
Entrance Lobby	NC 40
Pantry / Lavatories	NC 45 – 50

(*): In accordance to the requirements of HKBEAM Plus and LEED. Final NC field test shall be conducted by a HOKLAS-accredited independent agency, which shall be employed by the Contractor.

NC targets for spaces which have not been included in the above table should be consulted with the project acoustic consultant.

The design goals will be the lower of the range outlined above, with the upper level being a typically acceptable level.

2. Performance

- a) Maximum permissible sound levels in octave bands of airborne transmission through the combination of grille, registers, diffusers, terminal units, related pressure reducing devices and fan coil units, when operated in installed condition per plans and specifications, shall be no greater than the NC levels specified above.
- b) Equipment room noise levels and noise transmission to adjacent NSR (Noise Sensitive Receiver) shall comply with the requirement of Hong Kong Planning Standards and Guidelines, which shall be 5dBA below the requirement indicated in Environmental Protection Department of Hong Kong as follows

Time Period	HKEPD Acceptable Noise Level (ANL)	Final Design Target for HKPSG (ANL – 5 dBA) at NNSR*
Daytime (0700 - 2300)	70 dBA	65 dBA
Night-time (2300 – 0700)	60 dBA	55 dBA

(*): The check point is 1 m outside the window of the nearest noise sensitive receiver (NNSR). Correction will be added to the noise level recorded with tonal, impulsive and intermittence sound characteristic.

- c) The sound pressure level from vents or louvers shall be controlled in order to achieve 60 dBA or below at the position 3m from the vents or louver outlet to achieve the external noise criteria of 65 dBA (daytime) and 55 dBA (night-time).
- d) The total noise level resulting from the simultaneous operation of all systems, including regenerated noise, breakout noise, structure-borne noise shall meet the Local Noise Ordinance and HKBEAM Plus SS 5. Acoustic calculation or measurement report of the combined noise levels from all building service systems during daytime and night-time shall be submitted and endorsed by a Corporate Member of Hong Kong Institute of Acoustics (HKIOA) or equivalent.

- e) The maximum structural borne vibration levels that generated by all electrical and mechanical equipment including the pumps, AHUs, PAUs, fans, cooling towers, chillers, generators, lift as well as pipework and ductwork in RMS velocity, shall not exceed 400µm/s in any 1/3 octave frequency bands from 4Hz to 500Hz, for any one second average measured at any point of a concrete floor and wall within the occupied area. The measurement shall be performed and comply with the requirement of ISO 2631-2 “Mechanical vibration and shock – Evaluation of human exposure to whole body vibration – Part 2: Vibration in buildings (1Hz to 80Hz)”. The vibration measurement shall be conducted by an independent testing agency employ by the Contractor.

- f) Equipment sound power level

Equipment selections have a primary effect on the background noise. Equipment selected for operation near the point of maximum efficiency on the operating curve is the quietest for a given set of parameters.

Contractor shall submit all mechanical equipment noise data for approval, especially the noise level of Fans, for early review and assessment. When the equipment will be mounted over occupied spaces, necessary drywall enclosure shall be provided accordingly to meet the specified NC requirements.

PART 2 PRODUCTS

2.1 Acoustical Lining

Internal acoustical ductwork lining shall be:

1. Fibrous Glass
2. Facing for low velocity duct liner (max. 12 meters/sec).
 - a) Finish: Neoprene coated
 - b) Stenciled NFPA 90
3. Facing for circular ducts and medium/high velocity ductwork (over 12 meters/sec), or where indicated on drawings.
 - a) Finish: Perforated, minimum 28 percent open area, minimum 0.7mm thick galvanized steel.
4. Minimum Thickness.
 - a) In ductwork: 25 mm, unless otherwise noted on drawings.
 - b) In plenums: 50 mm.
 - c) In linear slot diffuser plenums: 13 mm.
5. Minimum Density
 - a) In ductwork: 32 kg per cubic meter.
 - b) In plenums: 48 kg per cubic meter.

- c) In linear slot diffuser plenums: 32 kg per cubic meter.
- 6. Fire properties: Comply with the local Fire Services Department's requirements.
- 7. Suitable for duct velocity of 30 meters per second, without erosion damage.
- 8. Dynamic loss coefficient: maximum 1.2.
- 9. Maximum thermal conductivity: 0.035 Watt/meter K, for 25mm thickness.
- 10. Noise reduction coefficient (NRC) for 25mm thick lining: minimum 0.75.
- 11. Base Bid Manufacturers:
 - a) Johns-Manville (Linacoustic)
 - b) Certainteed Products Corp.
 - c) Owens-Corning Fiberglas Corp.
 - d) Or Approved Equal.

2.3 Adhesive and Sealer

- 1. Fire Properties: To comply with local Fire Services Department's requirements.
- 2. In conformance with NFPA 90A, or equivalent international standards.
- 3. In conformance with BS476 Part 7 Class 1, or equivalent international standards.
- 4. Adhesive: similar to Benjamin Foster 81-99.
- 5. Sealer: similar to Benjamin Foster 82-07.

2.4 Non-Hardening Caulking

- 1. Guaranteed to be permanently elastic.
- 2. The caulking shall comply with the local Fire Services Department's requirements.
- 3. In conformance with BS476 Part 7 Class 1, or equivalent international standards.

2.5 Flexible Duct

The flexible ductwork shall be tested and installed in accordance with the Air Diffusion Council (ADC) "Flexible Duct Performance & Installation Standards." such as but not limited to, installed fully extended with radius elbows. All testing shall be in accordance with ASTM E-477 and/or ADC Test Code FD-72R1.

Flexible duct shall be sized for project friction loss design criteria, but not more than 2 m/s air velocity. The length of flexible duct connection shall be minimum 1.5 m.

Flexible duct shall be similar to Flexmaster U.S.A., Inc., Type 8 or 8M; Thermaflex, Type M-KE; or similar.

Acoustical performance of the flexible duct shall be as follows:

Flexible Ductwork Acoustical Performance in Accordance with ADC Test Code FD 72-R1 Attenuation - Insertion Loss (dBre:20Pa) for Straight 2 m length Zero Flow						
Octave Band Center Freq. (Hz)	125	250	500	1K	2K	4K
200 mm diameter	6	11	17	19	19	12
300 mm diameter	5	10	15	17	16	9

2.6 Diffuser/Grill acoustic performance

The diffuser should have a regenerated noise of 5 to 8 NC point lower than the designed NC level. If air is supply through the diffuser at light troffer diffuser, it shall be tested in the laboratory to evaluate the damper noise.

Maximum permissible sound power levels in octave bands of airborne transmission through the combination of grille, registers, diffusers, terminal units, related pressure reducing devices and fan coil units, when operated in installed condition per plans and specifications, shall be no greater than the maximum in the following table:

Maximum PWL re 10^{-12} Watts				
Octave Band	NC 30	NC 35	NC 40	NC 45
1	58	62	66	68
2	52	56	60	63
3	44	49	54	58
4	41	46	51	56
5	38	43	48	53
6	37	42	47	52
7	36	41	46	51
8	37	42	47	52

2.7 Pressure reducing box or fan powered box above ceiling radiated noise:

Maximum permissible radiated sound power levels in octave bands when operated over occupied spaces in an installed condition, above a ceiling, shall be as per the following table:

Maximum Radiated Sound Power (dB re: 10^{-12} Watt)				
Octave Band	NC 30	NC 35	NC 40	NC 45
1	68	72	76	79
2	56	70	74	77
3	57	61	65	68
4	55	60	64	68
5	52	57	62	68
6	51	56	60	65
7	60	65	70	75
8	60	65	70	75

2.8 VIUs

Attenuation of noise from VIU systems in NC 40 or below spaces relies to a large extent on the effectiveness of internal acoustic duct lining. Generally, the manufacturer data varied as much as 10 dB for high static model. Noise measurement to evaluate VIU generated noise shall be performed in mockup room (if any) or on site. Standard static model (with 30Pa or below) shall be selected. Also, at least 1m supply and return air duct shall be internally lined with 25 mm thick lining for NC 40 or below spaces.

2.9 Cooling Tower

Material for cooling tower silencers shall be of stainless steel. The sound pressure level of discharge and intake positions of cooling tower shall not be higher than the below values when it is at 75% load situation, measured at 15m away from cooling tower. Contractor shall submit acoustic calculation to ensure background noise at the NSRs due to the cooling towers (together with any necessary acoustic treatments) meet their corresponding acoustic requirements.

Sound Pressure Level (dB) in Octave Band Freq.								Location
63 Hz	125 Hz	250 Hz	500 Hz	1 kHz	2 kHz	4 kHz	8 kHz	
60	64	61	55	55	52	52	54	Discharge
68	63	56	54	52	49	48	48	Intake

3.0 Chiller

The maximum permissible sound pressure levels in octave bands of each chiller shall be 85dBA at 1m from chiller for all full and part load conditions in accordance with ARI standard 575, Method of measuring machinery sound within equipment rooms.

For controlling the external noise transmission from chiller plant, ventilation openings for chiller plant room shall be treated by silencer and dummy louver shall be blanked-off with metal sheet or drywall. Noise calculation shall be submitted to ensure background noise at the NSRs due to the chillers (together with any necessary acoustic treatments) meet their corresponding acoustic requirements.

3.1 Toilet exhaust fan

Internal duct lining/silencer shall be provided for toilet exhaust fan. Actual acoustic treatment shall subject to the selected equipment noise levels and shall be approved by the Project Manager and the project acoustic consultant. Individual exhaust air duct shall be provided, and single exhaust duct serving two adjacent rooms shall be avoided, otherwise a long cross-talk silencer shall be provided between air grills. The fan shall avoid placing between two duct elbows which may create turbulence noise.

3.2 Carpark Ventilation Fan and Pantry Exhaust Fan

Noise criteria for carpark and pantry shall meet overall sound levels of 60dBA or below for safety concerns (i.e. to allow someone to hear approaching people/vehicles in carpark and to provide clearly audible conversation environment in pantry area). Noise shall be controlled at the building exterior to comply with the local EPD standard and requirement as shown above.

3.3 E&M Plant Room

For controlling the internal noise transmission from E&M Plant Room (e.g. Transformer Room, Server Room) towards the occupied areas located adjacently, the maximum overall noise level of the selected equipment shall be controlled below 80dBA. Otherwise, acoustic enclosure shall be provided to reduce the breakout noise so that the NC criteria will be achieved. Equipment noise data shall be submitted for review and approval.

3.4 Motor Acoustical Performance

All Motor drives when installed per plans and specifications shall operate with noise level not exceeding 80dBA. Otherwise, necessary acoustic treatments shall be applied to meet the corresponding acoustic requirements. Noise levels shall be determined in accordance with IEEE Standard #85 "Test Procedure for Airborne Noise Measurements on Rotating Electric Equipment" or equivalent international standards.

3.5 Acoustic Silencers

1. Factory pre-fabricated.
2. Shell:
 - a) Galvanized steel not less than 1mm thick.
 - b) Leakproof at pressure differential of 2000 Pa.
3. Fire Properties of Media:
 - a) Fire properties shall be complied with local Fire Services Department's requirements.
 - b) Flamespread Classification: Not greater than 20, when tested in accordance with ASTM E84, NFPA Standard 255, or UL No. 723, or Class 1 in accordance with BS476 Part 7, or equivalent international standards.
 - c) Fire Propagation: in accordance with BS476 Part 6, the index of overall performance (AI) not exceeding 12, and sub-index (Ai₁) not exceeding 6, or equivalent international standards.
 - d) Ignitability: in accordance with BS476 Part 5, Class P, not easily ignitable, or equivalent international standards.
 - e) Fuel contributed and smoke developed: Not greater than 20, when tested in accordance with ASTM E84, NFPA Standard 255, UL No. 723 or equivalent international standards.
4. Media:
 - a) Minimum 48 kg per cubic meter density glass fiber or minimum 80kg per cubic meter density mineral fiber, packed under minimum 5 percent compression.
 - b) Filler to be inert, vermin proof, and moisture proof.
 - c) Where sound attenuators are selected for outdoor or connected to outdoor

behind weather louver, they shall be water-repellent and treated against corrosion. Test report issued by independent test laboratory for water vapour sorption of the sound absorptive infill shall be submitted for approval. The test result shall show that the amount of water vapor absorbed by the infill does not exceed 6 % by weight in accordance with ASTM C1104/C1104M-13a.

- d) Test report shall be submitted to show that the increase in weight due to water penetration into the silencer splitters shall be less than 0.1 kg by weight in one hour after the silencer is wet by water spraying.
- e) The Contractor shall provide laboratory test report for the fibreglass for this project to show that the airflow resistivity of fibreglass has achieved at least 23 kN.s/m⁴ in accordance with ISO 9053.
- f) The cooling tower silencer shall have dry film thickness of coating on exposed surface, when measured in accordance with BS 4842:1991, BS6496:1991, and shall not be less than 1.2 mil. Prepare and submit color samples to the Project Manager. Prepare 6 sets samples and gloss range and submit for approval.
- g) Galvanized perforated steel: Minimum 0.7 mm thick. Perforation not more than 30 % and hole diameter not larger than 2.5 mm so as to prevent any fibre erosion by the air stream.

5. Net Insertion Loss Ratings and Pressure Drop:

- a) Determined by duct-to-reverberant room test method at design airflow. The calculation for selection of sound attenuator shall be provided by the Contractor in accordance with the requirement of the particular specification.
- b) The maximum pressure drop across the selected attenuator shall be 50Pa, unless otherwise specified.

6. Certified Tests:

- a) Submit certified test data from approved laboratory for pressure drop and insertion loss ratings.
 - For square or rectangular attenuators: 600mm x 600mm.
 - For cylindrical attenuators: 600mm diameter.
 - For cone diffuser/silencers: Sizes indicated on drawings.
 - Certification data for pressure drop and net insertion loss: based on tests of same attenuator.
 - Attenuators and test reports: subject to inspection upon request of Architect.

7. Base Bid Manufacturers:

- a) Industrial Acoustics Company
- b) Kinetics Noise Control
- c) Nap Silentflo Or Approved Equal.

8. Acoustic Performance

Sound attenuators performance indicated below are guidelines only to meet NC and accepted noise level at NSR requirements. The Contractor shall employ their own acoustic specialist manufacturer to review the actual sound data of the proposed equipment such as AHUs, PAUs, EAFs, cooling towers, chillers, generators and other fans and define the sound attenuators performance to achieve the target NC and outdoor noise levels.

Acoustic calculations to predict the final NC or ambient noise levels shall be submitted by the acoustic specialist supplier to the Project Engineer for approval.

The dynamic insertion loss (dB) of the silencer, tested in accordance with ISO 7235, shall include but not limit to the following:

Silencer Application	Dynamic Insertion Loss (dB) in Octave Band Freq.								Approx. Length	Max. P.D.
	63 Hz	125 Hz	250 Hz	500 Hz	1 kHz	2 kHz	4 kHz	8 kHz		
For General External Noise Control	3	9	13	20	20	14	14	9	900 mm	50Pa
For NC 45 space or above	4	11	17	25	24	17	15	10	1200 mm	50Pa
For NC 40 Space or below	6	13	21	30	29	20	16	12	1500 mm	50Pa
For NC 35 Space or below	10	17	28	42	40	29	21	17	1800 mm	50Pa
For NC 30 Space or below	11	19	32	43	44	33	23	18	2100 mm	50Pa
For Ventilation Opening of Generators facing NSRs	12	27	42	52	65	67	39	26	2400 mm	50Pa
For Ventilation Opening of Generators not directly facing NSRs	11	24	39	49	59	61	37	24	2100 mm	50Pa
For Ventilation Opening of Chillers	7	13	20	36	40	36	27	16	1200 mm	50Pa
For Discharge of Cooling Towers	6	10	17	24	25	14	12	11	1500 mm	20Pa
For Intake of Cooling Towers facing NSRs	6	10	16	22	22	13	13	12	1200 mm	5Pa

3.6 Acoustic Louver

The acoustic louver shall be fabricated from galvanised sheet steel, aluminium or stainless steel sheet. The acoustic blades shall be filled with insert, non-combustible acoustic material. The acoustic louver shall meet the local Fire Services Department requirement. The acoustic and airflow performance shall be certified by independent testing laboratory.

1. Acoustic Performance:

Transmission Loss Tests shall be performed in accordance with ASTM E 90 or E336 or equivalent. The minimum thickness of acoustic louver shall be 300mm. The sound pressure level from louvers shall be controlled in order to achieve 60dBA or below at the position 3m from the louver outlet to meet external noise criteria. Acoustic louver performance indicated below are guidelines only to meet accepted noise level at NSR requirements. The Contractor shall employ their own acoustic specialist manufacturer to review the actual sound data of the proposed equipment and define the acoustic louver performance to achieve the target outdoor noise levels.

Transmission Loss Data (dB) in Octave Band Freq.								Approx. Thickness
63 Hz	125 Hz	250 Hz	500 Hz	1 kHz	2 kHz	4 kHz	8 kHz	
6	7	10	12	18	18	14	13	300mm

2. Base Bid Manufacturers:

- a) Industrial Acoustics Company
 - b) Kinetics Noise Control
 - c) Nap Silentflo
- Or Approved Equal.

3.7 Acoustic Enclosure

The acoustic enclosure shall be fabricated and installed in such a manner as to provide the acoustic performance, durability, and design requirements specified below. The acoustic enclosure manufacturer is fully responsible for the detailed design of the acoustic enclosures and the fixing details of the enclosure to the main structural steel work, purling including joint details between panels. Noise levels due to equipment noise breakout through acoustic enclosure shall permit attaining sound pressure levels in all 8 octave bands in occupied spaces conforming to the specified external noise criteria and NC levels.

2. Material:

The panels shall be constructed of galvanized steel of normally 1.6 mm thickness solid outside and 1.0 mm thickness perforated inside. Hole size shall not exceed 2.5 mm in diameter and account for 11 to 23% of the total surface area. Modules shall be welded, free draining, and free of cavities in which water may collect.

Acoustic infill material shall be of fiber acoustic material, non-corrosive, resistant to attack by fungus, fire-resistant, vermin proof, and non-hygroscopic. Fill material shall be free draining, self supporting and shall retain physical and sound absorptive characteristics after long term polymer sheeting. Acoustic infill as 100 mm thick 48 kg/m³; fiberglass (See enclosed Table 1 and Table 2).

3. Acoustic Performance:

Sound Absorption Coefficient Tests shall be performed in accordance with ASTM C 423, Type A mounting or with BS 3638 : 1963.

Transmission Loss Tests shall be performed in accordance with ASTM E 90 and ASTM E 413 or with BS 2750 Part 3 : 1980.

Table 1. Sound Transmission Loss Data – STC 40

Octave Band Centre Frequency, Hz	125	250	500	1K	2K	4K
Sound Transmission Loss Data, dB	21	29	39	50	56	56

Table 2. Sound Absorption Loss Coefficients – NRC 0.85

Octave Band Centre Frequency, Hz	125	250	500	1K	2K	4K
Sound Absorption Loss Coefficients	0.60	0.80	0.90	0.90	0.90	0.80

4. Access Doors:

Hinged doors openings shall be provided at suitable locations in the enclosure for access to the equipment for future maintenance. The construction of the door shall be similar as above and shall not jeopardize the acoustic characteristics of the entire enclosure. Door gasket shall be provided at door periphery to prevent any leakage.

5. Durability / Maintenance & Weathering Characteristics:

All exterior panel surfaces shall be protected from atmospheric corrosion.

All panel configurations shall be tested for corrosion resistance in accordance with ASTM B 117 in a NVLAP - certified or other equivalent international standards certified independent test laboratory accredited under the Accreditation Criteria Guide 25:1990 (E).

6. Fire Resistance:

Tests shall be performed on nonmetallic materials in accordance with ASTM E 84. All materials shall have a Class A fire rating with flame spread not greater than 15.

7. Method of Construction for Acoustic Enclosures

The acoustic enclosure shall be assembled by lowering consecutive panels between flanged steel beam posts with sound absorptive facing the noise source. Consecutive panels shall rest in one another in such a manner as to preclude line-of-sight gaps or openings through the panel.

Individual acoustic panels shall not be bolted, welded, or otherwise fastened to each other or to the structural support posts.

Individual acoustic panels shall be removable and replaceable. (It shall be shown that, should safety or other considerations mandate the use of mechanical equipment to raise and / or lower individual enclosures into position, the lift capacity of the equipment need not exceed 91kg).

Enclosure shall be installed such that enclosure surfaces facing both the noise source and the sensitive receiver are plumbed in accordance with accepted engineering specifications and to the satisfaction of the Project Manager.

8. Visual Appearance:

The acoustic panels shall be able to be installed with horizontal or vertical reveals to satisfy aesthetic considerations.

9. Documentation & Training:

Manufacturer to provide fully detailed operation and maintenance manuals including as built records drawings for wall acoustic panels including all fixing brackets and acoustic seals.

10. Base Bid Manufacturers:

- a) Industrial Acoustics Company
 - b) Kinetics Noise Control
 - c) Nap Silentflo
- Or Approved Equal.

3.8 Drywall Enclosure

1. Drywall Enclosure as specified shall be at least double layers of 15 mm gypsum board support on separated framework without any contact with the ductwork/silencer and silencer. Air spacing of 50 mm between ductwork/silencer surface and gypsum board shall be maintained and internally packed 25 mm thick insulation of minimum 48 kg/cum. All the gaps around acoustic enclosure shall be sealed up by acoustic sealant properly. Actual construction shall subject to the selected equipment noise levels and shall be approved by the Project Manager and the project acoustic consultant.
2. Fans/Silencers shall avoid hanging above the occupied spaces. Otherwise, Drywall Enclosure shall be provided and covering the fans and silencers, in order to reduce the equipment breakout noise and to meet the specified NC requirements.
3. Route main branch ducts in corridors only, with individual branches into rooms, in lieu of ducts crossing over noise sensitive rooms (e.g. office areas, conference room) and penetrating demising partitions. Otherwise, Drywall Enclosure shall be provided to reduce the duct breakout noise (i.e. to meet the specified NC requirements) and to minimize the noise transfer between connecting rooms via the air duct. Moreover, having the enclosure built would allow easier construction of partitions.

3.9 Duct/Pipe Lagging

1. Soil pipes, rainwater pipes, drainage pipes and main branch ducts shall avoid passing through the occupied spaces. Otherwise, Duct/Pipe Lagging shall be provided as specified. Lagging material shall be wrapped around 25 mm thick insulation and comply with the following requirement.

Surface Density	Min. 5 kg/m ²
Composite	• 1 layer of high density insulation core and, • 1 layer of strain proof foil facing on at least one side.

Fire	All composite layers shall comply to BS476 Part 4: Non-Combustible					
Operating Temperature	-30°C to +85°C					

2. The lagging material shall provide a minimum acoustic insulation as follows:

Octave Band Centre Frequency, Hz	125	250	500	1K	2K	4K
Transmission Loss in dB	5	10	15	27	30	24

3. The lagging material Acceptable Manufacturer: Subject to compliance with requirements, provide specified products of one of the following:

a) Euro-Soundstopper

b) Kinetics “KFA”

Or Approved Equal.

PART 3 EXECUTION

3.1 Acoustical Lining

- Adhere with 150 mm wide strips of adhesive.
 - 300 mm on centers.
 - At joints in lining.
- In addition, secure with weld pins and 50 mm diameter washers on 400 mm centers.
- Coat edges with sealer.
- Provide continuous sheet metal edge protectors at entering and leaving edges of lined duct sections, and all joints.
- Dimensions of lined ductwork are clear inside dimensions after lining has been installed.
- Extent of ductwork sound linings:
 - All VIUs supply and return duct for NC 40 or below spaces: 25mm thick.
 - Plenums at intake side and discharge side of AHUs/PAUs/EAFs: 50mm thick.
 - All toilet exhaust fans: 25mm thick.
 - Duct lining for VRF outdoor units if required: 25mm thick.
 - Transfer air ducts (if any): 25mm thick.
 - Where indicated on drawings.
- Where ductwork velocities exceed 12 meters per second, use of internal lining shall be reviewed by the Construction Manager in fulfilling the above requirements. If internal lining is deemed unsatisfactory for the particular application, provide perforated metal facing over internal lining, or sound traps as directed.

3.2 Sound Proof Construction for Duct Penetrations

1. Required for openings between ductwork and following construction:
 - a) Equipment room walls.
 - b) Floors, except in shafts.
2. Sound proofing
 - a) Fill openings with fibrous glass blanket or board for full depth of penetration.
 - b) Caulk each side of opening with non-hardening, non-aging caulking compound.

3.3 Air Distribution Systems

To achieve the least amount of system-generated noise and to assure containment of noise at or near its source. The following is the planning guideline that would influence the design of the mechanical systems.

- Configure air handler discharge ducts to be straight for at least three duct diameters from AHU. If a turn is necessary in the discharge duct, assure that it is in the same direction as the fan rotation to avoid generation of very low frequency duct rumble from reverse airflow.
- Provide significant runs of ductwork within Mechanical Equipment Rooms, secondary space and/or on the roof before entering occupied spaces. Ductwork shall have sufficient spacing to incorporate a silencer inside the AHU room and also 1 m duct length from fire damper.
- Route trunk and main branch ducts in corridors only, with individual branches into rooms, in lieu of ducts crossing over rooms and penetrating demising partitions. Where trunk or branch ducts must penetrate demising partitions and/or pass over sound sensitive spaces, either use round duct for its superior break-in/break-out characteristics, or use lagging to reduce breakout and cross-talk. Do not route exhaust ducts over sound sensitive spaces.
- Duct velocities shall be designed for laminar flow, since velocity generated turbulence creates noise. The following air flow velocity limits shall be used as a guideline for designing ductwork:

NC design level	Recommended air duct velocities (m/s)				
	riser/drop per shaft	main branch in ceiling in void	runout in diffuser	diffuser and diffuser plenum boxes	return/ extract grille
NC 30	9	6.5	4	1.8	2.2
NC 35	10	7	4.5	2	2.5
NC 40	10	7.5	5	2.5	3
NC 45	11	8	5	2.5	3
NC 50	12	8	5.5	3	3.5
external louvers	-	-	-	-	2.5

- Volume dampers are preferred at individual branch takeoffs in lieu of at the diffuser face. In general, attempt to locate dampers five or more duct diameters upstream of the diffuser, and assure that the duct between damper and diffuser is internally lined. If the individual branch duct from branch takeoff/damper to diffuser is flex, use insulated or acoustically rated duct.
- If there are volume control device, and additional 5-8 dB should be added based on the sound power levels.
- The diffuser should have a regenerated noise of 5 to 8 NC point lower than the designed NC level. If air is supply through the diffuser at light troffer diffuser, it shall be tested in the laboratory to evaluate the damper noise.
- Pressurized acoustically lined plenums with 50 to 100 mm thick acoustic liner shall be employed on the discharge or return to achieve significant reactive noise attenuation by the expansion chamber effect. Thicker acoustic lining provides improved low frequency attenuation.
- Near the Mechanical Equipment Rooms and where trunk ducts pass through the plenum over occupied spaces, round sheet metal duct is preferred for its noise containment quality, which is superior to rectangular or flat-oval duct. If round trunk ducts cannot be used, duct lagging or enclosure may be required in some sound sensitive spaces to minimize "breakout."
- Provide supply and return silencer for the AHUs/PAUs/EAFs. Silencer length subject to the selected sound power level of equipment and pressure drop shall be minimized to 50Pa. Sound attenuators performance indicated below are guidelines only to meet NC and accepted noise level at NSR requirements. The Contractor shall employ their own acoustic specialist manufacturer to review the actual sound data of the proposed equipment and define the sound attenuators performance to achieve the target NC and outdoor noise levels.
- Silencers and Fans shall be located within plant room area and not be located over occupied spaces. It can be installed at the vertical ductwork if necessary. Relocation shall be provided. Otherwise, drywall enclosure shall be provided and covering the acoustic silencers and fans. Acoustic treatment subjects to the sound power level of equipment.

3.4 Plumbing and Toilet

Some general noise control guidelines for plumbing are as follows:

- Cast iron type pipes are recommended, particularly for soil and rainwater pipes, which radiates less water running noise than those PVC pipes.
- Any soil pipes / rainwater pipes / drainage pipes run over the occupied areas with NC 40 or low background sound level shall be wrapped with 25 mm high density fiberglass with flexible lagging material wrap around the pipe to reduce the water flow noise.

- The lowest pressure possible in order to operate valves and provide the flow that desired for fixtures. Pipe velocities should be designed in accordance to the guidelines below with maximum 3 m/s.

Recommended Water Flow Velocities	
Pipe Diameter (mm)	Velocity Limit (m/s)
25 - 50	1.0
75	1.5
100	2.0
125	2.5

- Specify quiet devices such as quiet-type, flush valves and taps with full-ported nozzles and non-splash aerators. If toilet rooms are unavoidably next to sensitive spaces, use siphon-jet, tank-type water closets with adjustable flow valves.
- Install air chambers or shock-absorbing devices to prevent water hammer in lines subject to abrupt shut-off.

3.5 Sound Attenuators

1. All attenuators shall be fitted with end plates and packed in SEALED bags at factory prior to shipment.
2. Install in accordance with Manufacturer's Recommendations to obtain published performance.
3. Maximum Static Pressure Loss:
 - a) After installation: measure total system pressure before and after attenuators.
 - b) If pressure loss exceeds maximum static pressure loss schedules on drawings: at no charge, replace attenuators and/or modify entrance and/or discharge aerodynamic flow to obtain specified performance.
4. Provide sound attenuators for the following equipment:
 - a) All AHUs/PAUs/EAFs supply, return and external ventilation openings;
 - b) All ventilation fans discharge and intake;
 - c) All ventilation openings of chillers and generators;
 - d) All cooling towers discharge and intake.

3.6 Shop Drawings / Submittal

1. Catalog of sound attenuators and internal acoustic lining, including certified test data
2. Manufacturer noise data of all HVAC equipment
3. Acoustic calculation shows the resultant NC and/or ambient noise dBA level, including sound attenuator performance, static pressure loss, regenerated noise.

4. Acoustic treatment proposal, including silencer / louver schedule
5. Drawing indicates the acoustic treatment location, such as silencer and acoustic louver.
6. Drawing indicates the acoustic treatment location, such as the extent of internal acoustic lining, thickness and location.
7. Drawing indicates the drywall enclosure and acoustic enclosure details and their locations. (If applicable)
8. Certification that sound lining meets erosion test method described in UL Publication No. 181

VIBRATION CONTROL SPECIFICATION

PART 1 GENERAL

1.1 Description of Work

1. This section specifies vibration isolation for all mechanical, electrical and plumbing building systems.
2. The work in this section includes the following:
 - a) Vibration isolation devices
 - b) Equipment vibration balance
 - c) Equipment inertia bases
 - d) Thrust restraints and guides.
 - e) Piping and ductwork flexible connectors
3. Provide vibration isolation for all mechanical, electrical and plumbing building systems to prevent the transmission of vibration and mechanically transmitted sound to the building structure as indicated on drawings and as specified, complete.
4. Include adjustments of each mounting system, and the measurement of isolator system performance. Specific mounting arrangements for each item of mechanical equipment shall be as described herein, and as indicated by schedules and details on the Drawings.

1.2 Related Sections

1. Applicable provisions of the following codes and trade publications shall apply to the work performed under this section and shall be made part of the contract documents.
2. Relevant codes and trade standards include the following:
 - a) ISO 2372 and BS 4675 – Mechanical Vibration for Rotating Machinery
 - b) ANSI S2.19 – 1989 (R1987) – Mechanical Vibration - Balance Quality Requirements of Rigid Rotors
 - c) ANSI/AMCA Standard 204-96, - Balance Quality and Vibration Levels For Fans
 - d) NEBB, 1997 – Sound and Vibration in Environmental Systems
 - e) NEBB, 1997 – Procedural Standards for Measuring Sound and Vibration
 - f) ASHRAE Handbook, 2003 – HVAC Applications, Chapter 47 – Sound and Vibration Control
 - g) ARI Guideline G, 1995 – Mechanical Balance of Fans and Blowers
 - h) Local and National Building Codes, as applicable

1.3 Design Criteria

1. All mechanical, electrical and plumbing systems shall be vibration isolated as specified in the section.
2. Equipment vibration limits:

All equipment shall be statically and dynamically balanced to meet the following vibration limits under all design operating conditions and under support conditions comparable to the conditions specified in the equipment vibration isolation schedule, at the end of this section.

Equipment Type	Vibration Limit (mm/s, rms)
Pumps	2.5
Chillers	1.25
Cooling Towers	2.5
Direct Drive Fans	2.5
Belt Driven Fans	2.5
All other rotating Equipment	2.5
Reciprocating Equipment	10

- a) These vibration limits apply either on the bearings or the equipment support structure, whichever applicable.
- b) The vibration limits shall include the effects of inertia mass or inertia bases, where applicable.
- c) Equipment with variable frequency drives shall meet these limits throughout the entire frequency range that the equipment will operate.
- d) Equipment with Variable Frequency Drives (VFD) and supported on vibration isolation shall not be operated below the following rotational speeds:

Specified Vibration Isolation Minimum Static Deflection	Lower Limit of Rotational Speed (rpm)
Less than 25 mm	600
25 mm	500
50 mm	400
75 mm	350
100 mm	300
125 mm, or higher	250

The intent of this restriction is to prevent operation of the equipment in the range of rotational speeds that are near or within the range of resonant frequencies of the isolated system.

1.4 Submittals

1. Product Data, Shop Drawings, and Calculations:

- a) Complete descriptions of the products to be supplied, including catalog cuts, data sheets, model numbers, dimensions, specifications and installation instructions for all vibration isolation products.
- b) An itemized list of all isolated equipment with detailed schedules showing isolators proposed for each piece of equipment, referencing materials and drawings.
- c) The following list includes the required shop drawings that shall be submitted:
 - Equipment identification mark.
 - Manufacturer's model number for each vibration isolator, the equipment or ductwork or pipeline to which it is to be attached, and the number of isolators to be furnished for each installed system.
 - Show base construction for equipment; include dimensions, structural member sizes and support point locations.
 - For steel spring mounts or hangers - free height, deflected height, solid height, isolator loading, and diameter of spring coil.
 - For neoprene isolators - free height, deflected height, and isolator loading.
 - Dimensional and weight data for concrete inertia bases, steel and rail bases, and details of isolator attachment.
 - Furnish calculations to meet applicable codes for all equipment, isolated or non-isolated, piping and ductwork.
- d) Show methods of support for ceiling suspended isolated equipment.
- e) Detail methods of vibration isolation for ducts, pipes and trays penetrating walls and slabs.
- f) If requested, submit samples of vibration isolation devices to be used for this project. After review and approval, samples will be returned and can be installed at the job site. All costs to supply and return such samples shall be borne by the Contractor.
- g) Provide installation instructions, drawings and field supervision to insure proper installation and performance of systems.
- h) Upon completion, submit written certification from equipment manufacturer that vibration isolation is installed correctly and properly adjusted.

1.5 Quality Assurance

1. Manufacturers to supply products specified in this section shall have a minimum of 10 years documented relevant experience.
2. Companies contracted to install products specified in this section shall have a minimum of 10 years documented relevant experience.
3. Vibration isolators shall have markings or other means of determining actual deflection after installation and adjustments have been completed.

4. Vibration isolators shall operate within the linear portion of their load versus deflection curves.
5. Neoprene isolators shall have a shore hardness of 30 to 50, within +/- 5%, after minimum aging of 20 days.
6. Substitution or deviation from the vibration isolation specified in this section shall be acceptable as long as the equipment manufacturer provides written guarantees that specified noise and vibration limits shall be met. Any costs to remedy noncompliant conditions shall be fully borne by the Contractor.

1.6 Contractor's Responsibility

1. The Contractor shall have the following responsibilities:
 - a) Determine vibration isolation sizes and locations.
 - b) Provide equipment vibration isolation as specified.
 - c) Guarantee specified isolation system deflections, to give a total transmissibility of not more than 10% in slab on ground and not more than 5% in upper levels.
 - d) Provide calculations by a registered professional engineer in mechanical or building services discipline (experienced in the field of noise and vibration) and endorsed by a member of Hong Kong Institute of Acoustic, substantiating all systems meet applicable code requirements.
 - e) Provide installation instructions, drawings and field supervision to insure proper installation and performance of systems.
2. The Contractor shall be responsible for rebalancing, realigning, other remedial required actions, including replacement of any systems, that are found to produce noise and vibration exceeding manufacturer's and project's requirements.
3. The Contractor shall be responsible for all costs associated with remedial work to correct deficient installations or installations that generate undue noise and vibration.
4. Existing Conditions: The Contractor shall notify the Architect of any site conditions, which adversely affect vibration isolation system installation or performance. Installation shall be suspended until approval is received.

PART 2 PRODUCTS

2.1 General

1. All vibration isolation described in this section shall be the product of a single manufacturer. The products of specific manufacturers are identified to establish type and quality.
2. Approved manufacturers:
 - a) Mason Industries
 - b) Amber Booth Co.

c) Kinetics Noise Control

d) or Approved Equal

Products of other manufacturers shall be acceptable provided they comply with specifications and are pre-approved by The Project Engineer.

3. Vibration isolators shall be provided in accordance with the weight distribution to produce uniform deflection. Furnish deflections indicated.
4. Where specific type of vibration isolation is not shown or specified, provide isolators recommended by the isolation manufacturer compatible with equipment arrangements shown.
5. A single manufacturer for all vibration isolation equipment is required, unless specifically approved in writing by the Architect.

2.2 Materials

1. Vibration Isolators

a) General

- Spring diameters shall be no less than 0.8 of the compressed height of the spring at rated load.
- Springs shall have a minimum additional travel to solid, equal to 50% of the rated deflection.
- Springs shall have a horizontal stiffness at least 100% of the vertical stiffness to assure stability.
- Corrosion-resistant when located in a corrosive environment.

b) Type FS - Free Standing Spring

- Spring isolators shall be freestanding, and laterally stable without any housing and complete with minimum 6 mm thick non-skid neoprene pads between the base plate and the support.
- Spring elements shall be selected to provide static deflections as shown on the vibration isolation schedule.
- All mountings shall have leveling bolts, rigidly bolted to the equipment.
- Mountings shall be hot dipped galvanized for corrosion resistance.
- Isolators shall be Amber Booth Type SW, Kinetics Type FDS, Mason Type SLF, or equal.

c) Type RS - Restrained Spring

- Restrained spring isolators shall be used for equipment subject to load variations and large external forces.
- Isolators shall consist of large diameter, laterally stable, steel springs assembled

into welded steel housing assemblies designed to limit movement of the supported equipment in all directions.

- Housing shall include vertical resilient limit stops to prevent spring extension when weight is removed, as when equipment is drained.
- Spring elements shall be selected to provide static deflections as shown on the vibration isolation schedule.
- A minimum clearance of 12 mm shall be maintained around restraining bolts and between the housing and the spring so as not to interfere with the spring action.
- Limit stops shall be out of contact during normal operation.
- Mountings shall be hot dipped galvanized for corrosion resistance.
- Isolators shall be Kinetics Type FLS, Mason Type SLR, or equal.

d) Type HS - Spring Hanger

- Vibration hanger shall contain a steel spring located in a neoprene cup manufactured with a grommet to prevent short-circuiting of the hanger rod.
- Minimum 12mm clearance between retainer box and spring hanger rod.
- The hanger bracket shall be designed to carry a 500% overload without failure and to allow a support rod misalignment through a 30° arc without metal-to metal contact or other short circuit.
- Isolators shall be Kinetics Type SH, Mason Type 30, or equal.

e) Type NM – Neoprene-In-Shear Floor Mount

- Double deflection neoprene-in-shear mounts shall have a minimum static deflection as specified in the vibration isolation schedule but no less than 7.5 mm.
- All metal surfaces shall be neoprene covered to prevent corrosion and have friction pads both top and bottom so they need not be bolted to the floor.
- Bolt holes shall be furnished for those areas where bolting is required. Top and bottom surfaces shall have non-skid ribs
- Not greater than 70 durometer, maximum compression set at 25% of the original deflection.
- Isolators shall be amber Booth Type RVD, Kinetics Type RD, Mason Type ND, or equal.

f) Type HR - Resilient Hanger

- Vibration hanger shall consist of a double deflection neoprene-in-shear element encased in a welded steel bracket, not exceeding a static deflection of 10 mm.
- Steel retainer box encasing neoprene mounting.
- Isolators shall be Amber Booth Type BRD, Kinetics Type RH, Mason Type HD,

or equal.

g) Type NP - Neoprene Pad

- Isolation pads shall be single ribbed or crossed, double ribbed elastomers, in combination with steel shims where required.
- Pads shall be sized to have minimum static deflections as specified in the vibration isolation schedule.
- Suitable top bearing plate provided to uniformly distribute load.
- 2mm galvanized steel plate between multiple layers of pad thickness.
- Pads shall be 40 to 60 durometer. Include neoprene shoulder washers to prevent short-circuiting due to through bolting.
- Isolators shall be Amber Booth Type NR, Kinetics Type NP and NGS, Mason Type W and SW, or equal.

h) Type PM –Pad type elastomer mountings to incorporate the following:

- High quality bridge bearing neoprene.
- Maximum loading 7,000 KPa.
- Suitable bearing plate to distribute load.
- Minimum thickness, 50mm.
- Mason Type BBP or as approved.

i) Type HRS – Resilient Spring Hanger

- Vibration hanger shall contain a steel spring in a neoprene cup and minimum 30 mm thick neoprene in-shear element, in series.
- The hanger bracket shall be allowed a support rod misalignment through a 30° arc without metal-to metal contact or other short circuit.
- Isolators shall be Kinetics Type SRH, Mason Type 30N, or equal.

j) Type HPS, Pre-compressed Spring Hanger

- Vibration hanger shall be as specified for Type HRS or HS, but shall be pre-compressed to the rated deflection so as to keep the piping or equipment at a fixed elevation during installation.
- The hangers shall be designed with a release mechanism to free the spring after the installation is complete and the hanger is subjected to its full load.
- Deflection shall be clearly indicated by means of a measuring device.
- Isolators shall be Mason PC30N, Mason PC30, or equal.

2. EQUIPMENT BASES AND RAILS

a) General

- All Curbs and roof rails are to be bolted or welded to the building steel or anchored to the concrete deck for resisting a minimum 110 MPH wind loads.

b) Type SR - Steel Rails

- Rails shall be for equipment having legs or frames that do not require a complete steel base.
- Members shall be rigid to prevent stress in the equipment, and shall be cross-braced as required by manufacturer to prevent twisting under design loads.
- Rails shall be constructed of welded structural steel beam, angle or channel members.
- Provisions for height saving brackets to reduce operating height, as required.
- Rails shall be Mason Type R or ICS, Kinetics Type SBB, or equal.

c) Type ISB - Integral Structural Steel Base

- The base shall be reinforced as required to limit base flexure and equipment misalignment to equipment manufacturer specifications.
- All perimeter steel members shall have a minimum depth equal to 1/10 of the longest dimension of the base.
- Base depth shall not exceed 300 mm provided that base flexure and equipment misalignment are kept within the limits specified by the manufacturer.
- Height saving brackets shall be employed in all mounting locations, as required, to provide a minimum base clearance of 50 mm.
- Bases shall be Mason Type M or WF, Kinetics type SFB, or equal.

d) Type ICB – Inertia Concrete Base

- Provide floating concrete bases where shown on drawings.
- Bases for split case pumps shall be large enough to provide support for suction and discharge elbows.
- Base depth shall be a minimum of 1/12 of the longest dimension of the base, but not less than 150 mm.
- Base depth shall not exceed 300 mm unless specifically recommended by the base manufacturer for mass or rigidity.
- Forms shall include minimum concrete reinforcing consisting of 12.5 mm diameter reinforcing bars welded in place 150 mm on-center, running both ways in a layer 38 mm above the bottom of the base.
- Forms shall be furnished with steel templates for anchor bolt sleeves and equipment attachments.

- Height saving brackets shall be employed in all mounting locations to maintain 50 mm clearance below the base.
 - Bases shall be Amber Booth Type CPF, Kinetics Types CIB-H and CIB-L, Mason Type K, or equal.
- e) Type SRC - Spring Roof Curb
- Vibration isolation curbs shall be complete assemblies designed to resiliently support the equipment at the specified elevation and shall constitute a fully enclosed air- and weather-tight system.
 - Steel springs to be laterally stable, provided with rust resistant finish and rest on 6mm thick neoprene pads.
 - Spring elements shall be selected to provide static deflections as shown on the vibration isolation schedule.
 - All spring locations shall have removable waterproof covers to allow for spring adjustment and/or removal.
 - The roof curb shall be built to restrain the rooftop unit. Calculations shall be provided and stamped by a professional licensed engineer with at least 5 years of design experience.
 - Roof curbs shall be Mason Type RSC or approved equal.

3. Thrust Restraints

- a) Horizontal thrust restraints shall consist of a spring element seated in a steel washer reinforced neoprene cup at the bottom, in series with a molded neoprene element.
- b) Steel springs shall be free standing and laterally stable. Spring diameters shall be no less than 0.8 of the compressed height of the spring at rated load. Springs shall have a minimum additional travel to solid equal to 50% of the rated deflection.
- c) The spring element shall be designed so it can be preset for thrust at the factory and adjusted in the field to allow for a maximum of 6 mm movement at start and stop.
- d) The assembly shall be furnished with rod and angle brackets for attachment to both the equipment and the ductwork or the equipment and the structure.
- e) Horizontal restraints shall be attached at the centerline of thrust and symmetrical on either side of the unit.
- f) Horizontal thrust restraints shall be Mason Type WBI/WBD, or equal.

4. Flexible Connections

- a) Flexible connections include duct, piping and electrical connections to vibration-isolated equipment.
- b) Flexible neoprene connectors shall be used to connect all piping to isolated

equipment, except equipment for which flexible connectors are not permitted by code.

- c) Flexible duct connections shall be fabricated from materials that are compatible with the intended service for a particular system.
- d) Flexible duct connections shall result in a loose and resilient connection and maintain a minimum clearance of 100 mm between the two sides that they connect.
- e) Flexible pipe connectors shall be one of the following two kinds:
 - Hose type rubber and metal connectors with minimum specified live lengths.
 - Expansion joint, stainless steel bellows without braided jackets or twin spherical rubber type.
 - Hose type flexible connectors shall have minimum live lengths as specified in the 2003 ASHRAE Handbook, Table 43 (see table below).

Nominal Diameter (mm)	Minimum Live Length (mm)
25	300
50	300
75	450
100	450
125	600
150	600
200	600
250	600
300	900

5. Pipe expansion joints

- a) They shall be either stainless steel bellows without braided jackets or twin spherical rubber connectors.
- b) They shall be selected for the design operating condition and shall comply with manufacturer requirements.
- c) Where control rods or cables are required they shall be installed not in tension and shall include thick neoprene washers to prevent metal-to-metal contact.
- d) Expansion joint connectors shall be Mason Type SAFEFLEX, or approved equal.

6. Acoustical Sleeves

- a) Where piping or ductwork passes through equipment walls, floors or ceilings, provide a split acoustical seal with minimum 25 mm closed cell neoprene sponge bonded to the inner faces.
- b) Once the pipe or duct is inserted through the seal, the seal shall be tightened to eliminate clearance between the inner sponge face and the penetrating element.
- c) Concrete shall be packed around the seal to make it integral with the floor, wall or ceiling if such sleeve is not already in place around the pipe prior to the construction of the building member.

- d) Where temperatures exceed 240°F., 10 lb. density fiberglass shall be used in lieu of the sponge.
- e) Seal shall be Mason Type SWS, Mason Type SPS, or equal.

PART 3 EXECUTION

3.1 General

1. Install isolation systems in strict accordance with the manufacturer's written instructions, applicable building codes, and approved submittals.
2. Floor mount and resilient hangers of the same type can be used interchangeably as long as they are selected to provide the specified minimum static deflection and meet all product specification requirements.
3. Flexible connectors shall be used to connect all piping and ductwork to isolated equipment, except equipment for which flexible connectors are not permitted by code.
4. Coordinate work with other trades to avoid rigid contact with the building. Inform other trades following work, such as plastering or electrical, to avoid any contact which would reduce the vibration isolation.
5. Provide restraints for equipment per applicable code. Design and provide restraints to prevent permanent displacement in any direction caused by lateral motion, overturning, or uplift. Restraint of equipment must not inhibit vibration isolation.
6. Housed spring mounts shall not be permitted unless specifically approved by the Architect.
7. Vibration isolators shall not cause any change of position of equipment resulting in stress on equipment connections.
8. All vibration isolators must be sized according to weight, distribution, static and dynamic loads to be supported. Isolator ratings should be expressed as static deflection based on the actual load supported.
9. Piping and ductwork: When determining vibration isolation requirements, use either single pipe/duct diameter or equivalent pipe/duct diameter for multiple pipes/ducts supported together on racks, whichever is applicable.
10. Piping and ductwork penetrations through walls and floors shall not make rigid contact with the building structure. Sleeves shall be provided to allow for contact free penetrations. Any such penetrations shall be smoke-proofed and fire-stopped in an approved manner.
11. Replace isolators which do not produce the required deflection, are incorrectly loaded, or do not produce the required isolation as approved and at no extra cost to the Owner.
12. All drains, fire sprinkler piping, gas lines, vacuum lines and emergency exhaust ductwork shall not require vibration isolation regardless of location and size.

3.2 Installation

1. Allow 150mm minimum clearance between any vibrating equipment and nearby building structures.
2. Allow 500 mm minimum unobstructed space in the ceiling areas for the installation of vibration isolation hangers.
3. Allow adequate clearance for vibration isolators on all mechanical equipment.
4. All equipment bases shall be placed in position and supported temporarily by blocks or shims prior to the installation of the equipment isolators.
5. Spring isolators shall be installed after all equipment is installed without changing equipment elevations.
6. Thrust restraints shall be installed on all cabinet fan heads, axial or centrifugal fans whose thrust exceeds 10% of unit weight.
7. All roof-mounted isolators shall either be bolted or welded to the building steel or anchored to the concrete deck for resisting code and design wind loads.
8. Resilient hangers shall be installed as near as possible to the supporting overhead structure. The equipment suspension points shall be located in a rigid and heavy portion of the building structure. Suspension of equipment from lightweight floor slabs is not allowed.
9. Suspension rods shall be attached to rigid members of the equipment structure. When such attachment points do not exist, a heavy steel framework shall be provided to support the equipment with suspension rods attached to this framework.
10. Electrical connections to vibration-isolated equipment shall be made with long lengths of flexible steel conduit or flexible armored cable. These flexible connections shall be located so as to prevent rigid connections between the resiliently mounted equipment and the building structure.
11. Drain connections from isolated equipment to floor drains shall have at least 25 mm air gap above the drain.
12. All debris shall be removed from beneath the equipment. The Contractor shall verify that there are no short circuits of the isolators or the isolation system. The equipment shall be free to move in all directions.
13. After the entire installation is complete and under full operational load, the spring isolators shall be adjusted so that the load is transferred from the blocks to the isolators.

3.3 Piping and Ducting

1. Install flexible duct connectors on all pipes and ducts connected to vibration-isolated equipment.
2. Flexible duct connections at fans and air handling units shall have 100 mm clear gap between collars.

3. Pipes and ductwork connected to vibration-isolated equipment shall be supported on isolation of the same type and with static deflection equal to the equipment isolators for a minimum horizontal distance of 15 meters from the equipment.
4. All pipes 300 mm or greater in diameter or pipe racks with equivalent pipe diameter greater than 300 mm, such as cooling towers and chilled water supply and return pipes, shall be spring-isolated within 15 meters (or 100 diameters if greater than 15 meters) from connected rotating equipment. Size springs to provide a minimum static deflection of 25 mm (Type HS or Type HPS).
5. Electrical connections to equipment, which is supported or suspended by vibration isolators, shall be made with long lengths of flexible steel conduit or flexible armored cable.

3.4 Installation

On completion of installation of all vibration isolation devices herein specified, the Owner's representative shall inspect the completed system and report in writing any installation error, improperly selected isolation devices, or other faults in the system that could affect the performance of the system. A written report shall be submitted outlining corrective work necessary to comply with the above specifications. Corrective work shall be the responsibility of the E&M Contractor.

3.5 Vibration Isolation Schedule

Equipment Types	Base Type	Isolator Type	Static Defl (mm)	Vibration Isolator Type
Fans for motor size 5kW or larger (including AHU, PAU, Axial In-Line Fan) Floor/Roof Supported below - 300 rpm (or VFD) 301 - 450 rpm 451 - 600 rpm 601 - 750 rpm 751 - 850 rpm 851 - greater Suspended	per Mfr (1)	Spring Sprg/Neop	100 90 75 50 38 25 (above+6)	FS (e.g Mason SLF/SLR) HRS (e.g. Mason 30N)
Fans (Propeller)	per Mfr (1)	Neoprene	2	NP (eg. Mason ND)
Pumps 5HP - greater	ICB (e.g Mason KSL)	Spring	50	RS (eg. Mason SLR)
Pumps - less than 5HP	ISB (e.g Mason MND)	Neoprene	10	NP (eg. Mason ND)
Associated Equipment (Assoc. w/ equipment & piping/ductwork)	per Mfr (1)	(2)	(2)	(2)
Piping / Ductwork Suspended Floor Supported	See Specification	Spring	30 25	HS (e.g. Mason 30) RS (eg. Mason SLR)

Equipment Types	Base Type	Isolator Type	Static Defl (mm)	Vibration Isolator Type
Transformers Floor Mounted	per Mfr (1)	Neoprene	5	NP (e.g Mason W)
Chillers	per Mfr (1)	Restr. Spring	75	RS (e.g Mason SLR)
Generators	per Mfr (1)	Restr. Spring	50	RS (e.g Mason SLR)
Cooling Towers	per Mfr (1)	Restr. Spring	75	RS (e.g Mason SLR)
Heat Exchangers	per Mfr (1)	Restr. Spring	50	RS (e.g Mason SLR)
Indoor / Outdoor Condensing Unit	per Mfr (1)	Neoprene	5	NP (e.g Mason W)

(1) Base as recommended and/or provided by equipment manufacturer.

(2) Same as connected pump/equipment.

(3) FS: Free Stand Spring

HRS: Resilient Spring Hanger

ICB: Inertia Concrete Base

ISB: Integral Structural Steel Base

HS: Spring Hanger

RS: Restrained Spring

NP: Neoprene Pad

END OF SECTION